

SRM IST DELHI NCR CAMPUS, MODINAGAR
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DATA SCIENCE (21CSS303T)

3ND YEAR, 6TH SEMESTER

SESSION: 2025-2026



ASSIGNMENT-4

SUBMITTED BY:

DURGESH NANDINI
RA2311030030046

SUBMITTED TO:

Ms. GEETANJALI TYAGI

Q Explain the concept of data wrangling & its importance.

Ans: It is a process of organizing, cleaning and transforming raw data into a structured format so it can be used for analysis. In real world scenarios, raw data collected from different sources is often incomplete, inconsistent or unorganized.

Importance of Data Wrangling:

- ① Improve the quality and reliability of data.
- ② Helps in removing errors, inconsistent and duplicates.
- ③ Makes data structure and ready for analysis.
- ④ Saves time during analysis and modeling.

In data science a large portion of time is spent on data wrangling high-quality leads to better insights and predictions.

Q Explain string manipulation in python with suitable code examples.

Ans: String manipulation refers to the process of modifying, analyzing or processing text data using various string function available in python.

Since many datasets contain text information such as names, addresses or messages, strings manipulation is an important part of data processing.

Python provides many building methods for working with strings such as changing case, searching, replacing, slicing and concatenation.

< code >

```
text = "Hello World"
print(text.lower())
print(text.upper())
print(text.replace("World", "python"))
print(text[0:5])
name = "Durgesh"
greeting = "Hello" + name
print(greeting)
```

string manipulation helps in clearing and processing textual data before analysis.

③ Explain the general techniques and programming practices used for efficient handling of large datasets in data science.

ans: Handling large dataset efficiently is an important challenge in data science because large amount of data require more memory, processing, processing power and time. Therefore, several techniques and practices are used to process large datasets effectively.

Common techniques:

- ① Data sampling: Analyse a small representative portion of data.
- ② Chunk Processing: Read data in small parts instead of location of entire dataset.
- ③ Using efficient libraries: Tools like Numpy and Pandas for faster processing.
- ④ Parallel Processing: use multiple CPU cores to process data faster.

Compression: Reduce storage and memory usage. ③

④ Explain different data transformation techniques used in preprocessing.

ans: Data transformation is a preprocessing step where raw data is converted into a suitable format for analysis or machine learning algorithm. This step ensure that the data is consistent, standardized and meaningful.

⑤ Common data transformation techniques include:

① Normalization: Scaling values b/w 0 & 1.

② Standardization: Adjusting data to have mean 0 and standard deviation 1.

③ encoding: converting categorical data ~~ex~~ into numerical form.

④ Aggregation: Combination multiple values into summary statistics.

⑤ Smoothing: Removing noise from data.